Examiner only

6. Two students used modelling to illustrate the effect of camouflage in predator-prey relationships. In their investigation, spaghetti was used to model the prey and forceps to model the predator. They used the following method.

Method

1. Mark out a 1 m^2 area of green grass on a field.
2. Mix together 20 pieces of green spaghetti and 20 of red spaghetti, all 5 cm in length, in a beaker.
3. Empty the contents of the beaker in the marked area, ensure all the pieces are evenly spread out.
4. Use a pair of forceps to pick up as many pieces of spaghetti as possible in 30 seconds. Remove these from the model.
5. Count the number of spaghetti pieces left on the grass after 30 seconds.
6. For each pair of a colour left on the grass, add one more piece of spaghetti of the same colour to model the process of reproduction.
7. Repeat steps 4 to 6 twice more.

The results of the experiment are shown in **Table 6**.

Table 6

	Number of pieces of spaghetti present on the grass		Number of pieces of spaghetti added	
	Green	Red	Green	Red
At start	20	20		
After 1st pick	18	10	9	5
After 2nd pick	22	8	11	4
After 3rd pick	27	5		

(a) (i) Suggest what would happen to the number of red spaghetti pieces present on the grass, after a 4th pick. [1]

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(ii) State the name of the evolutionary process being modelled in this investigation. [1]

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(b) State the type of reproduction represented in this model. [1]

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(c) This model has limitations. Suggest **two** limitations in this model. [2]

Limitation 1 ....

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Limitation 2 ....

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5

Turn over.